2026 Spring

Course ID: 1142-1764

Foundations of Computer Science

01-Course Introduction

1

HOUR

10:10 – 11:00

Topic

Course Introduction

Cloud Computing

Course Introduction

Welcome to the course — let's get started.

Week 1

50 minutes

Course Info

Introduction to Computer Science

Fundamentals · Organization · Networks · OS · Algorithms · Data Structures · ADT

2 Semesters · 11 Topics · 32 Weeks · 3 Credits each

INSTRUCTOR	DEPARTMENT	LANGUAGE	TOOLS
Prof. H.T. Chu	**International Program in Computer Science**	**English (EMI)**	**AWS Academy**

ASSESSMENT (EACH SEMESTER)

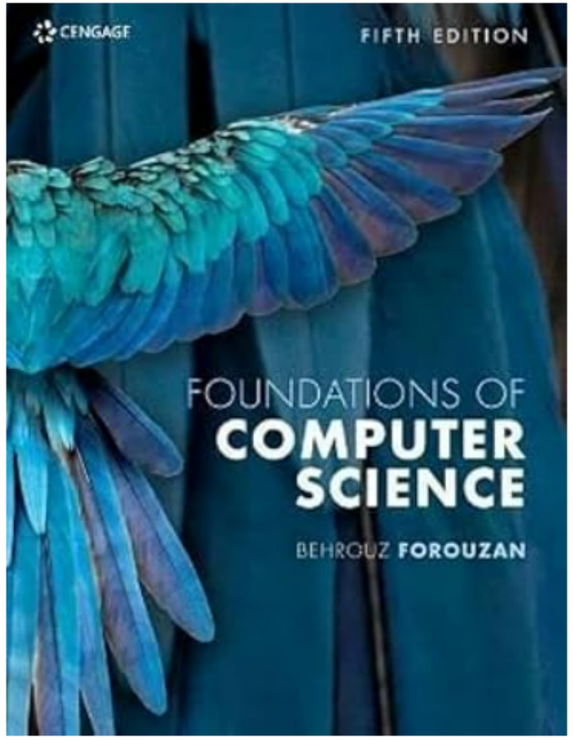
Assignments	Midterm Exam	Project / Labs	Final Exam	Participation
30%	**20%**	**30%**	**20%**	**Bonus 5%**

Providence University · Department of Computer Science and Information Engineering · AY 2025–2026

Thursday, 3-4 (10:10-12:00)

Providence Hall R307

The Textbook



Foundations of Computer Science Paperback

作者 Behrouz Forouzan (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

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語言	尺寸	ISBN-10	ISBN-13
English	18.9 x 24.7 x 2.6 cm	1473787327	978-1473787322

查看所有詳細內容

1 Introduction

Turing Model
Von Neumann Model
Computer Components
History
Social and Ethical Issues
Computer Science as a Discipline
Outline of the course
End-Chapter Materials
Practice Set

2 Number Systems

Introduction
Positional Number Systems
Non-Positional Number Systems
End-Chapter Materials
Practice Set

3 Data Storage

Data Types
Storing Numbers
Storing Text
Storing Audio
Storing Images
Storing Video
End-Chapter Materials
Practice Set

4 Operations on Data

Logic Operations
Shift Operations
Arithmetic Operations
End-Chapter Materials
Practice Set

5 Computer Organization

Introduction
Central Processing Unit
Main Memory
Input/Output Subsystem
Subsystem Interconnection
Program Execution
Different Architectures
A Simple Computer
End-Chapter Materials
Practice Set

6 Computer Networks

Introduction
TCP/IP Protocol Suite
Layers
Internet Applications
The World Wide Web (www)
End-Chapter Materials
Practice Set

7 Operating Systems

Introduction
Evolution
Components
A Survey of Operating Systems
End-Chapter Materials
Practice Set

8 Algorithms

Concept
Three constructs
Algorithm Representation
A More Formal Definition
Basic Algorithms
Subalgorithms
Recursion
End-Chapter Materials
Practice Set

9 Efficiency of Algorithms

Definition
Common Algorithms
Big-O Notation
End-Chapter Materials
Practice Set

10 Programming Languages

Evolution
Translation
Programming Paradigms
Common Concepts
End-Chapter Materials
Practice Set

11 Software Engineering

The Software Life cycle
Analysis Phase
Design Phase
Implementation Phase
Testing Phase
Documentation
End-Chapter Materials
Practice Set

12 Data Structure

Arrays
Records
Linked Lists
End-Chapter Materials
Practice Set

13 Abstract Data Types

Background
Stacks
Queues
General Linear Lists
Trees
Graphs
End-Chapter Materials
Practice Set

This semester

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+ aws academy

Number Systems

Binary & Hexadecimal

How computers speak — and how we read them

0110 1000 0101 1100 1010 0x1F 0xA3 0xFF

Computer Science Fundamentals

Why Do Computers Use Different Number Systems?

01

Hardware Reality

Transistors have two states: ON or OFF.
This maps directly to 1 and 0.

↔

Human Readability

Hexadecimal condenses long binary
strings into compact notation.

∞

Universal Standard

Memory, colors, and protocols all use hex in computing.

Decimal (base-10) is natural for humans. Binary (base-2) is natural for electronics. Hex (base-16) bridges the two worlds.

Binary — Base-2 Number System

Key Concepts

- Only digits: 0 and 1
- Each position = a power of 2
- Rightmost = $2^0 = 1$
- Next left = $2^1 = 2$
- Then $2^2 = 4$, $2^3 = 8$ ...
- 8-bit group = 1 byte

Example: $1010\ 1101_2 = 173_{10}$

1	0	1	0	1	1	0	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

$128 + 0 + 32 + 0 + 8 + 4 + 0 + 1 = 173$

Low nibble: $1101 = D_{16}$

High nibble: $1010 = A_{16}$

💡 A 4-bit group is called a nibble. Two nibbles = 1 byte (8 bits).
Binary 1010 1101 = Hex 0xAD = Decimal 173

Hexadecimal — Base-16 Number System

Key Concepts

- 16 possible digits per position
- Uses 0–9 then A–F
- A=10, B=11 … F=15
- Prefix 0x (e.g. 0xFF)
- Each hex digit = 4 bits
- 2 hex digits = 1 byte
- Compact: 0xFF vs 11111111

Hexadecimal Digit Table

0	1	2	3	4	5	6	7
0	1	2	3	4	5	6	7
8	9	A	B	C	D	E	F
8	9	10	11	12	13	14	15

decimal →

Example: 0x2F

$2 \times 16^1 + F \times 16^0$

$= 2 \times 16 + 15 \times 1 = 32 + 15 = 47_{10}$

Decimal ⟷ Binary ⟷ Hexadecimal

Decimal	Binary	Hexadecimal
0	0000	0x0
1	0001	0x1
5	0101	0x5
10	0000 1010	0x0A
15	0000 1111	0x0F
16	0001 0000	0x10
42	0010 1010	0x2A
100	0110 0100	0x64
255	1111 1111	0xFF

💡 **Tip:** 0xFF = 255 (max byte) | Split 4 bits → 1 hex digit | Powers of 2 appear at 0x1, 0x2, 0x4, 0x8, 0x10…

Real-World Applications

Web Colors

`#FF5733`

CSS color codes are 3 hex bytes
(R, G, B — each 00–FF)

Memory Addresses

`0x7FFF0020`

Pointers in C/C++ and
debugger output use hex

ASCII / Unicode

`'A' = 0x41`

Character encodings are
stored as hex values

File Magic Bytes

`FF D8 FF E0`

JPEG files always start
with these hex bytes

Network (MAC/IP)

`AC:DE:48:00`

MAC addresses use
hex-separated pairs

Assembly / Debug

`MOV EAX,0x1`

Register values and opcodes
are shown in hex

Summary

Decimal (Base-10)

What humans use daily. Digits 0–9.

Binary (Base-2)

What computers use internally. Digits 0–1.
1 byte = 8 bits = values 0–255.

Hexadecimal (Base-16)

Human-readable shorthand for binary.
Digits 0–9, A–F.
2 hex digits = 1 byte.

Quick Reference

```
1 bit          = 0 or 1
4 bits         = 1 nibble = 1 hex digit
8 bits         = 1 byte = 2 hex digits
0b prefix      = binary  (0b1010)
0x prefix      = hex     (0xFF)
255₁₀          = 1111 1111₂ = 0xFF₁₆
16₁₀           = 0001 0000₂ = 0x10₁₆
```

Master these three systems and you'll understand how computers store, address, and process all information.

2

HOUR

11:10 – 12:00

Topics

- Computer
- Cloud Computing

Computer & Cloud Computing

Computer

Hardware
CPU, memory, storage, I/O devices — how a computer works

Software
OS, applications, system architecture layers

Networking
Connecting computers: LAN, WAN, protocols

Evolution
From mainframes → PCs → mobile → cloud

Cloud Computing

What is Cloud?
On-demand computing resources delivered over the internet

Service Models
IaaS, PaaS, SaaS — what each layer provides

Deploy Types
Public, Private, Hybrid — overview

Key Providers
AWS, Microsoft Azure, Google Cloud Platform

Computer Basics	Cloud Overview	Live Demo	Q&A
20 min	20 min	10 min	5 min

Cloud Computing

Deployment Models

3 Core Deployment Types

01	Public Cloud
02	Private Cloud
03	Hybrid Cloud
+	NaaS — Network as a Service

01 Public Cloud

Shared infrastructure over the internet

Third-party cloud providers (AWS, Azure, GCP) own and operate computing infrastructure, delivering services to multiple customers over the internet on a shared, pay-as-you-go basis.

Scalability

Instantly scale up/down to match demand — no hardware planning

Cost Model

Pay-per-use OpEx; zero upfront CapEx required

Maintenance

Provider fully manages updates, patches, and hardware

Availability

99.9%+ global SLA with built-in redundancy

USE CASES

- E-commerce & web apps
- SaaS platforms
- Dev / Test environments
- Big data & analytics
- Content delivery (CDN)

Major Providers: AWS · Microsoft Azure · Google Cloud Platform · IBM Cloud · Oracle Cloud

02 Private Cloud

Dedicated infrastructure for one organization

Infrastructure exclusively provisioned and operated for a single organization — on-premises or hosted by a third party — providing full control, enhanced security, and regulatory compliance.

Control

- Full admin access
- Custom hardware configs
- Tailored network topology
- Proprietary software stack

Security

- Isolated network perimeter
- GDPR / HIPAA compliance
- Data sovereignty
- Air-gapped environments

Performance

- Predictable low latency
- No noisy neighbors
- Dedicated resource pools
- Custom SLA terms

Common sectors: Banking · Healthcare · Government · Defense · Research Institutions

03 Hybrid Cloud

Best of private + public combined

A hybrid cloud integrates private and public cloud environments through secure connectivity, allowing workloads to move between them based on cost, compliance, and performance needs.

PRIVATE CLOUD

- Sensitive data
- Core systems
- Compliance workloads
- Low-latency operations

🔗 *VPN · Direct Connect · API Gateway*

PUBLIC CLOUD

- Burst & peak capacity
- Dev & test workloads
- Big data processing
- Disaster recovery

Key benefits: Agility + Control · Cost optimization · Workload portability · Gradual migration path

NaaS *Network as a Service*

NaaS delivers network infrastructure — bandwidth, routing, firewalls, VPNs, and WAN connectivity — as a cloud-based subscription service. Organizations consume networking capabilities without owning or managing physical equipment.

On-Demand Bandwidth

Scale network capacity instantly — no hardware provisioning cycles

Virtual Firewall & SD-WAN

Deploy security policies and WAN routing entirely via software

Pay-as-you-Use

OpEx subscription model — pay only for what you consume

Centralized Management

Manage your entire network fabric from a unified dashboard

Providers: Cisco Meraki · VMware VeloCloud · Aryaka · Zscaler · Cato Networks · AT&T NetBond

Summary

Comparing all deployment models

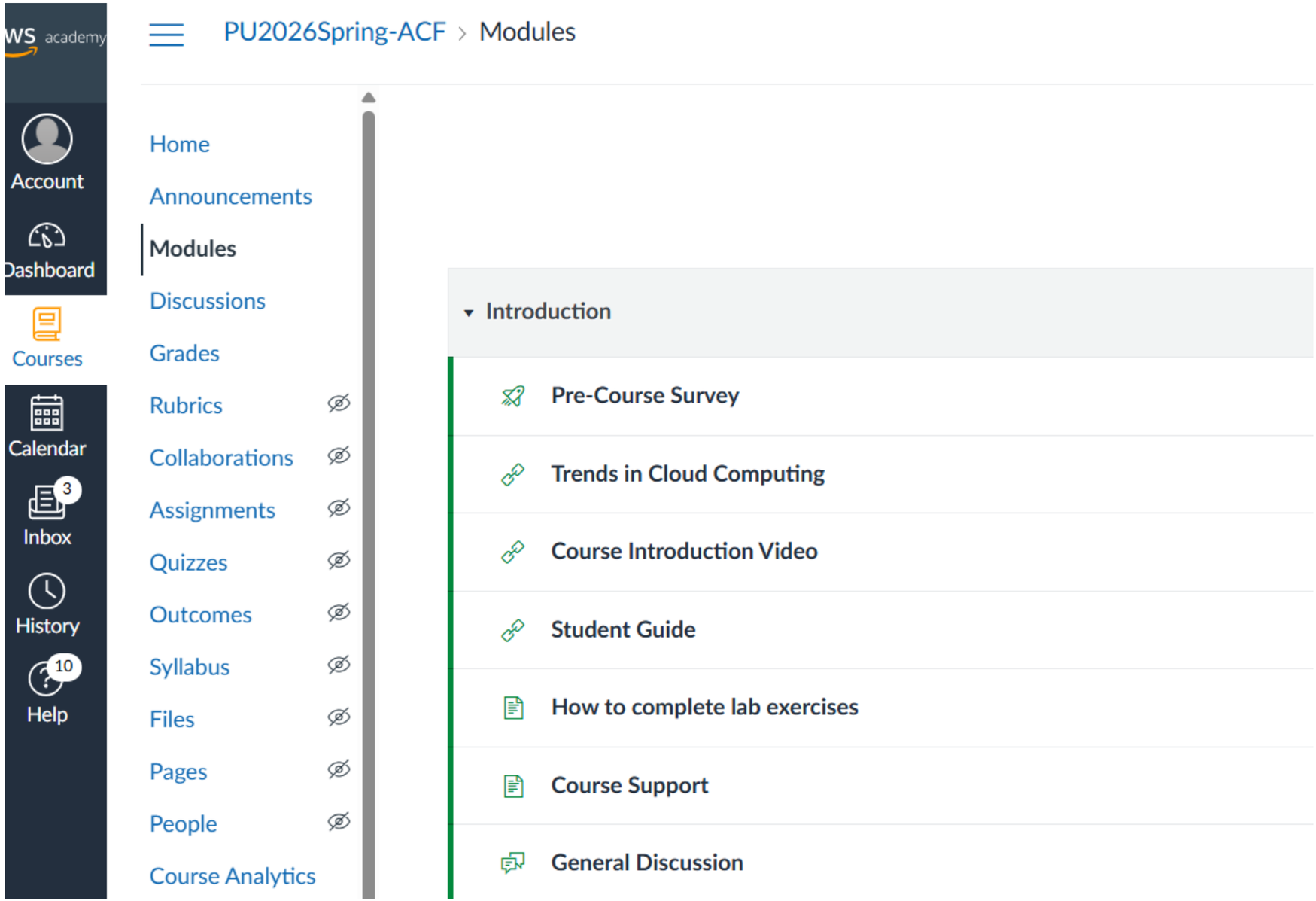
Model	Ownership	Scalability	Security	Cost Model
Public	Provider	★★★★★	★★★	Pay-per-use
Private	Enterprise	★★★	★★★★★	High CapEx
Hybrid	Both	★★★★	★★★★	Balanced
NaaS	Provider	★★★★★	★★★★	Subscription

AWS Academy Courses

Courses

Hidden Courses

MY ACTIVE CLASSES

Course	Nickname	Course Code	Start Date	End Date
AWS Academy Cloud Architecting [160615]	PU2026Spring-ACA	ACAv3EN-US-LTI13-160615	02/22/2026	07/21/2026
AWS Academy Cloud Developing [160614]	PU2026Spring-ACD	ACDv2EN-LTI13-160614	02/22/2026	07/21/2026
AWS Academy Cloud Foundations [160613]	PU2026Spring-ACF	ACFv2EN-LTI13-160613	02/22/2026	07/21/2026

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